Connecting EXHALE to the Lower Atmosphere: Survey, Implementation, and Example Applications

Stale note (2026-10-05): the coupling is no longer one-way in energy: `element_flux_closure.py` *with* `energy_coupling` *passes the heat the wind conducts into the column (*`base_conductive_flux_cgs`*) to* `photochem_to_lower_profile.py` `--conducted-heat-flux`*, which solves the column energy balance below the match (*`md/Update_EXHALE_stage3.md` *sections 76-78); the radiation half (the transmitted XUV handed down) is still not implemented.*

1. Motivation and current state

EXHALE's lower boundary is the $\sim$1 μbar level: the input "Planet radius" is the *radius of that level* (user-guessed), the base temperature is fixed at $T_0 = T_{\rm eq}$, the base density n_0 is an input (or pressure-anchored via `base_p_ubar`), and the gas is purely atomic H/He + trace metals at prescribed abundances. No molecules exist anywhere in the domain. The Wind-AE IC port carries a *simplified* molecular layer (μ-blend `molec_adjust` + bolometric heating/cooling, auto-shutoff when the atomic transition reaches the base).

[2026-08-15: this paragraph describes the tree as of 2026-07 and is no longer current. The coupled molecular network ($H_2/H_2^+/H_3^+/HeH^+$, `Molecular chemistry: True`*), H_3^+ cooling, Lyman–Werner photodissociation, the analytic lower column and the* `base.inp` *handoff are all implemented in* `src/modules/lower_atmosphere/`*. They default off, so an ordinary run is still the atomic base described here; the status table of each tier is the authoritative account.]*

This memo (i) surveys how the literature connects escape models to the lower atmosphere, (ii) documents the implementation in `src/modules/lower_atmosphere/` and `src/utils/run_lower.py`, (iii) applies it to the four planets modeled with EXHALE (HD 209458 b, HD 189733 b, WASP-121 b, WASP-52 b), and (iv) prescribes how the outputs are used and connected to EXHALE runs.

2. What the literature does at the interface

Key quantitative facts:

- **Base insensitivity for strong winds** (Salz et al. 2016, §3.5–3.6): $T_{\rm base} \pm 40\% \Rightarrow \dot{M} \pm 5$–$10\%$ (linear); base density $10^{14} \rightarrow 10^{12}$ cm^{-3} $\Rightarrow \dot{M} \lesssim 40\%$ lower. Their error budget: base density $\pm 50\%$, base T $\pm 10\%$, both dwarfed by the XUV uncertainty ($\times 3$). $\Rightarrow$ *EXHALE's fixed-$T_{\rm eq}$ base is already defensible for hot Jupiters.*

- **Where the fixed base breaks:** (i) planets near radiative equilibrium (stable thermospheres); (ii) cool/sub-Neptune planets where H_2 survives into the wind, Salz's GJ 1214 b test: H_2 10–25% throughout, H^- a significant coolant, $\dot{M} - 15\%$; Taylor et al. (2026): adding H_2 *halves* the He 10830 signal (14%$\rightarrow$7%); Koskinen et al. (2022) hot Uranus: base $q_{H_2} \approx 0.84$ and H_3^+ IR emission is the dominant radiative coolant below $\sim 2.3\,R_{\rm p}$.

- **What a photochemical lower model uniquely provides** (Lavvas et al. 2014; Lavvas & Arfaux 2021): the $H_2 \rightarrow$H transition can sit exactly at 1 μbar and is temperature-sensitive there; photochemistry (OH-catalyzed H_2 destruction) dissociates H_2 even at 500–1000 K; *atomic-metal release profiles* (Na, K, Mg, Ca, Fe, Al, Si

Approach	Lower atmosphere	Hands to the wind model at $\sim$1 μbar	Cost / availability
Salz et al. (2016), TPCI	none: CLOUDY solves the whole column (molecules disabled)	n/a (single domain; base at n = 10^{14} cm^{-3}, $T_{\rm eq}$)	$\sim 3 \times 10^5$ CPU-h for 18 planets; interface not public
Koskinen et al. (2022)	*analytic*: isothermal-$T_{\rm eq}$ hypsometric column from 1 bar, $\mu(p)$ from chemical-equilibrium H_2/H/He fits (Visscher et al. 2006)	base *radius* r_0(1 μbar), T_0=$T_{\rm eq}$, $q_{\rm H_2}/q_{\rm H}/q_{\rm He}$; base v from mass conservation	trivial (closed-form; all equations in the paper)
Huang et al. (2017)	*analytic spacer*: isothermal μ=2.3 column, 1 bar $\rightarrow$ 10 μbar	base radius of the atomic layer; Lyα *absorbing* bottom boundary (H_2 accidental resonances, τ=1 at $N_{\rm H_2} \approx 10^{14}$ cm^{-2})	trivial
Taylor et al. (2025, 2026)	full Lavvas & Arfaux (2021) model (RC temperature + $\sim$150-species kinetics of Lavvas et al. 2014 + haze microphysics; $10^3 \rightarrow 10^{-6}$ bar)	T(1 μbar), altitude of the 1 μbar level, composition (with a *tolerated discontinuity*), K_{zz}	in-house, **not public**; coupling is a manual splice, iterated $\sim$3$\times$

return to atomic form below $\sim 10^{-3}$ bar), the physically-motivated base abundances for EXHALE's trace metals; photoionization-dominated electron density ($\sim 10^8$ cm^{-3}, 100$\times$ Saha); radical/haze heating of $\pm$100–400 K right at the handoff level. *(2026-08-19: EXHALE carries no O/OH/H_2O species of its own, so this chemistry reaches the code only through the handoff.)*

- **The published coupling is loose, not monolithic:** Lavvas et al. (2014) take $T(p < 1\,\mu\text{bar})$ *from* the Koskinen thermosphere and use "species $<$ 3 amu escape at the wind velocity" as their upper BC; Taylor et al. tolerate a composition discontinuity at 1 μbar. Only TPCI runs a tight coupling at every timestep (and it disables molecules).

3. The three-tier plan

Tier 1: analytic lower column (Koskinen-2022 style), deriving the 1-μbar base radius and base H_2/H/He from a closed-form column. **Tier 2**: molecular extension of EXHALE (H_2, H_2^+, H_3^+, HeH^+ chemistry + H_3^+ IR cooling), required only for warm Neptunes / sub-Neptunes; **Tier 2a**: an interim EOS-only molecular-base correction. **Tier 3**: consume a real lower-atmosphere model (photochemistry + radiative–convective $T(p)$) through a file interface; the open-source `VULCAN` (Tsai et al. 2017, 2021) + `HELIOS` (Malik et al. 2017) stack is the practical replacement for the non-public Lavvas & Arfaux code, and a TPCI-style full coupling is rejected (cost, availability, and demonstrated base insensitivity). Which science case needs which tier: hot Jupiters $\rightarrow$ Tier 1 only; He 10830/Hα work $\rightarrow$ Tier 1; metal transmission lines $\rightarrow$ Tier 3 (atomic-metal release abundances); warm Neptunes / sub-Neptunes $\rightarrow$ Tier 2 + 1.

4. Implementation

All Fortran lives in `src/modules/lower_atmosphere/`; the Python driver in `src/utils/`. Everything is **opt-in / default off**: with no keys and no `base.inp`, standard runs are unchanged by the lower-atmosphere code (the molecular chemistry off-switch reproduces the atomic systems byte-for-byte; the absolute production baseline itself moved with the 2026-07-23 defaults: staged secondary ionization, `He_rec_coupling`, He–H charge exchange).

4.1 Tier 1: `lower_column.f90`

Equations. The isothermal hypsometric relation is integrated from the 1-bar level (radius $R_{1\mathrm{bar}}$, in practice the broad-band transit radius) to the base pressure $p_0 = 1\,\mu$bar,

$$\frac{\mathrm{d}\ln p}{\mathrm{d}r} = -\frac{\mu(p,T)\,m_\mathrm{H}\,g(r)}{k_\mathrm{B}T}, \qquad g(r) = \frac{GM_\mathrm{p}}{r^2}, \quad T = T_\mathrm{eq}, \tag{1}$$

with a fixed-step 4th-order Runge–Kutta in $x = \ln p$ (2000 steps; the integrand is smooth). The mean molecular weight follows the chemical-equilibrium H_2/H partition of Visscher et al. (2006) as fitted by Koskinen et al. (2022, Eq. 11),

$$q_{\mathrm{H_2}} \simeq \frac{1.9845 + 10^u - \sqrt{10^u\,(3.9690 + 10^u)}}{2.3670}, \qquad u = -\frac{23672}{T} - \log_{10} p_{[\mathrm{bar}]} + 6.2645. \tag{2}$$

An implementation subtlety uncovered during validation: Eq. (2) returns the *full-mixture* volume mixing ratio $q_{\mathrm{H_2}} = n_{\mathrm{H_2}}/(n_{\mathrm{H_2}} + n_\mathrm{H} + n_\mathrm{He})$ (a protosolar He abundance is baked into the fit), so the fraction x_2 of H *nuclei* bound in H_2 must be recovered as

$$x_2 = \frac{2\,q_{\mathrm{H_2}}\,(1 + f_\mathrm{He})}{1 + q_{\mathrm{H_2}}} \le 1, \qquad \mu = \frac{1 + 4 f_\mathrm{He}}{(1 - x_2) + x_2/2 + f_\mathrm{He}}\ [\mathrm{amu}], \tag{3}$$

with $f_\mathrm{He} = n_\mathrm{He}/n_{\mathrm{H(nuclei)}}$ the elemental ratio (EXHALE's `HeH`). Equation (3) was verified against the Koskinen et al. (2022) Model A base composition ($q_{\mathrm{H_2}} = 0.84$, $q_\mathrm{H} = 0.026 \Rightarrow x_2 = 0.985$); the naive H-subsystem inversion $x_2 = 2q/(1+q)$ is *wrong* and misses the gate.

Interface. `lower_column_solve(Mp, R_deep, T, fhe, p_deep, p_base, r_base, qh2_b, qh_b, qhe_b, mu_b, r_base_atomic)` (cgs; pressures in bar). Because chemical equilibrium *underestimates* H dissociation at $T_\mathrm{eq} \approx$ 1000–2000 K (photochemistry ignored; Moses et al. 2011; Koskinen et al. 2013a), the routine also integrates the same column with the fully atomic $\mu = (1 + 4f_\mathrm{He})/(1 + f_\mathrm{He})$ and returns `r_base_atomic`: the two radii *bracket* the truth. The public helper `q_h2_equilibrium(p,T)` exposes Eq. (2) (NaN-safe, clamped to $[0, 1]$).

EXHALE key. `Lower column: <R_1bar in R_J>` in `input.inp` triggers a startup report: the derived $r_0(1\,\mu\mathrm{bar})$ (equilibrium + atomic bracket), the input "Planet radius" for comparison, and the base $q_{\mathrm{H_2}}/q_\mathrm{H}/q_\mathrm{He}/\mu$, plus a warning when the equilibrium base is strongly molecular. *Report only: nothing is overridden* (overriding is Tier 3's job, §4.4).

Validation gate (passed). Koskinen et al. (2022) Model A (Uranus-mass planet, $M = 8.6813 \times 10^{28}$ g, $R_{1\mathrm{bar}} = 2.5559 \times 10^9$ cm, Sun-like star at 0.05 au, $T_\mathrm{eq} = 1140$ K, $f_\mathrm{He} = 0.0793$):

	$r_0/R_{1\mathrm{bar}}$	$q_{\mathrm{H_2}}$(base)	q_H(base)
this module	1.343	0.838	0.027
Koskinen et al. (2022)	1.34	≈0.84	≈0.026

Approximations (documented in the source): isothermal column at $T_{\rm eq}$; spherical gravity (the Roche-equipotential correction matters only near RLO); chemical equilibrium only.

4.2 Tier 2a: the `Molecular base` key

`Molecular base: True` applies an *EOS-only* correction at startup: the H nuclei bound into H_2 at $(1\,\mu\mathrm{bar}, T_0)$ per Eq. (2)–(3) are removed from the base particle budget,

$$\texttt{ntot_bc} \;\rightarrow\; \texttt{ntot_bc} - \frac{x_2/2}{1+f_{\rm He}}, \tag{4}$$

which lowers the base pressure ($p_{\rm base} = \texttt{ntot_bc}\, n_0 k_{\rm B} T_0$) and raises the base mean molecular weight, i.e. a heavier, more compact base, the leading-order effect of a molecular base on the hydrodynamics. The species arrays remain atomic (H_2 chemistry is Tier-2 proper), so this is a *crude, clearly-flagged* approximation. *[2026-08-15:* `Molecular chemistry: True` *turns this key on by itself (an atomic* `ntot_bc` *beneath a molecular base is physically inconsistent), so it need not be set separately for a Tier-2 run.]* Example: HD 209458 b, $q_{\rm H_2}(1\,\mu\mathrm{bar}, 1450\,\mathrm{K}) = 0.831 \Rightarrow \texttt{ntot_bc}\,1.0 \rightarrow 0.546$.

4.3 Tier 2 foundation: `h3p_cooling.f90` and `mol_rates.f90`

H_3^+ infrared cooling. Optically thin cooling per molecule. The LTE emission $W(T)$ of one molecule into one steradian [W molecule^{-1} sr^{-1}] is the cooling function of the MiZATeP line list (Mizus et al. 2017, MNRAS 468, 1717, section 5 and Table 7; `references/Mizus_2017MNRAS_468_1717.pdf`), tabulated at 1 K steps from 20 to 5000 K in the ExoMol file `1H3_p__MiZATeP.cf` and carried as the module `h3p_mizatep_cooling_table` (generator `src/utils/h3p_mizatep_cooling_table.py`). Between the nodes $\ln W$ is linear in $1/T$ (a local Boltzmann factor, monotone); below 20 K the first interval's Boltzmann form is continued, so $W \rightarrow 0$ as $T \rightarrow 0$; above 5000 K the 5000 K value is held (H_3^+ is destroyed well below). The volumetric rate is

$$\Lambda_{H_3^+} = n_{H_3^+}\, 4\pi\, W(T)\, s_{\rm nonLTE}(T, n_{H_2}) \times 10^7 \quad [\mathrm{erg\,s^{-1}\,cm^{-3}}], \tag{5}$$

where $s_{\rm nonLTE}$ is the departure factor of Miller et al. (2013, Table 6; their Oka & Epp detailed-balance approach; `references/Miller_2013_JPCA_117_9770.pdf`), a ratio of two vibrational-level sums normalized by its own LTE sum, so it multiplies the MiZATeP W unchanged. It is interpolated as a power law in n_{H_2} between the columns $n_{H_2} = 10^6$–10^{14} cm^{-3} and linearly in T between the rows $T = 300$–5000 K; below 10^6 cm^{-3} the collisional limit $s = s(T, 10^6)\, n_{H_2}/10^6$ is used, above 10^{14} cm^{-3} the table edge is held (`h3p_nonlte_factor`). The tabulated values are upper limits per the paper's caveat on the proton-hopping rate. Both interpolators use the NaN-safe `.not.(x>edge)` bracketing adopted after the 2026-07-02 code review. **Why not the Miller et al. (2013) LTE fit.** Their Table 5 fits, used here until 2026-10-10 (update log stage 3 section 132), lie above MiZATeP by factors 1.10, 1.48, 1.89 and 2.1 at 900, 1500, 2000 and 3000 K; the NMT line list they state they summed, summed in LTE with their own partition function, agrees with MiZATeP to 1% over 300–2000 K, so the fit is the outlier (`md/h3p_cooling_cause_20261010.md`; header of `h3p_cooling.f90`). From 50 to 700 K the two agree to 4%. **Validation:** the built module returns $W = 2.92879 \times 10^{-20}$, 1.29793×10^{-19}, 3.19812×10^{-19}, 9.54546×10^{-19} W sr^{-1} at 1000, 1500, 2000, 3000 K, the ExoMol values (update log stage 3 section 132).

Molecular reaction rates. `mol_rates.f90` provides the full $H_2/H_2^+/H_3^+/HeH^+$ thermochemical network of Koskinen et al. (2022, Table 1; transcribed and verified against the PDF page): recombination (R1–R2, Storey & Hummer 1995), Voronov (1997) collisional ionization (R3–R4), $H_2^+/H_3^+/HeH^+$ dissociative recombination (R5–R7, R16; Larsson et al. 2008), the H_3^+ production backbone $H_2^+ + H_2 \rightarrow H_3^+ + H$ (R8, 2×10^{-9}), thermal and electron-impact H_2 dissociation (R12, R14), three-body chemistry (R13, R15), the HeH^+ chain (R17–R20), and H/He charge exchange (R21–R23). The stellar photo-rates P1–P5 (including H_2 dissociative and double photoionization) follow EXHALE's existing `PH_heat` routines once H_2 cross sections are added. Lyman–Werner neutral photodissociation is not part of the source network (Koskinen et al.'s sensitivity test bounded its effect on $\dot{M}$ at $\times 1.4$); EXHALE adds it separately and opt-in, keyed on the band flux at the planet (`Stellar LW flux`, 2026-08-13, §9.).

Tier-2 core (implemented 2026-07-03). The coupled system `System_HeH_mol.f90` solves 7–8 unknowns per cell (H^+, He^+, He^{++}, optionally He 2^3S, plus the H-nucleus fractions bound in H_2, H_2^+, H_3^+, HeH^+) as element-conserving steady-state balances with `hybrd1`: the atomic rows use EXHALE's own rate arrays (so the molecule-free limit reproduces the atomic systems' solution) and the molecular channels are the Table-1 network via `mol_rates` (R21/R22 H–He charge exchange deliberately excluded to preserve the atomic limit). The H_2 photoionization cross section (Yan et al. 1998, Eqs. 17–19; verified against their Table 7 to $<0.2\%$) enters the opacity, the P_{H_2} photoionization integral, the photoelectric heating, and the absorbed-energy normalization through optional arguments of `PH_heat_HHe`; H_3^+ IR cooling enters the `cool` array. The EOS counts each molecule as one particle, each molecular ion's electron, and the correct masses (`calc_ne/ntot/rho`, `get_species_densities`). `f_sp` grows to 37 columns (output schema and Load-IC names extended). Key: `Molecular chemistry: True` (requires He; v1 excludes trace metals; the molecular base particle count is coupled on automatically, an atomic `ntot_bc` under a molecular base is physically inconsistent). **Gates passed** (numbers re-established 2026-07-16 after the deterministic chemical-equilibrium seeding of the molecular solve; the exact gate inputs are pinned at `lower_atmosphere_figs/data_g*/input.inp`): (0) molecules-off regression, molecular chemistry off reproduces the atomic systems; (1) HD 209458 b molecules-on: a fully molecular base ($x_{H_2} \simeq 0.996$) with a sharp $H_2 \rightarrow$H front at $r = 1.020\,R_p$ and an essentially atomic wind above it (the hot-Jupiter atomic base is now a *result*, not an assumption), while below the front the molecular base suppresses He 2^3S by orders of magnitude (He 2^3S+H_2 Penning destruction, $\sim 10^4\times$ at the front, rising back to the atomic value above $\sim 1.3\,R_p$); (2) a hot-Uranus-like case ($0.0457\,M_J$, $R_p = 0.49\,R_J$, T_{eq}=1140 K, HD 209458 b orbit and spectrum): the front sits at $r = 1.156\,R_p$ and H_3^+ is active below it (peak $\sim 7\times 10^4$ cm^{-3} at $r \approx 1.04$). The $H_2 \rightarrow$H fronts and molecular base composition are essentially unchanged from the 2026-07-16 gates (front 1.019→1.020, 1.166→1.156), confirming the dissociation-front result is robust to the 2026-07-23 production defaults; refreshed 2026-07-23 at a shared 12000-step relaxation-snapshot convention (both HD209 runs $\log \dot{M} = 10.58$; relaxation snapshots, not flux-flat converged). Figure 1 shows the gate runs. Both gate configurations were Newton-converged on 2026-08-13 once the H_3^+ cooling entered the marching energy update; the converged solutions differ substantially from these snapshots below the $H_2 \rightarrow$H front and are given in §7..

He 2^3S + H_2 Penning ionization. When `Include He23S?` is also on, the metastable couples to H_2 through Penning ionization He(2^3S)+$H_2 \rightarrow$He(1^1S)+$H_2^+ + e^-$: the dominant He(2^3S) loss toward an H_2-dominated base (García Muñoz

2025, A&A 698, A199, Fig. 4). Its rate coefficient `penning_HeI23S_H2` (`Cool_coeff.f90`) is an analytic fit to García Muñoz Table A.5 (Cohen & Lane 1977 cross sections), $k(T) = 5.3791 \times 10^{-12}\, T^{0.6760}\, e^{-695.21/T}$ cm^3 s^{-1}, reproducing the tabulated 500–10000 K points to $\leq 0.13\%$. It enters `System_HeH_mol` as an H_2 loss / H_2^+ source / He(2^3S) sink and adds $(E[2^3S] - \mathrm{IP}[H_2]) \approx 4.4$ eV of electron heating; the minor associative H+HeH$^+$ branch ($\sim$10%) is folded into the Penning channel.

The _adv post-process is atomic-only, so the Penning-H_2 term does not appear there. The advection-corrected reconstruction in `post_process_adv.f90` (header, lines 4–13) treats the gas as H/He + trace metals and excludes the molecular species entirely: the molecular densities are never passed in or re-solved, and its `calc_ne`/`calc_ntot` calls omit the `nmol` argument. Consequences for a molecular run: (i) the _adv n_{tot}/n_e omit the neutral-H_2 particle count and the molecular-ion electrons; (ii) the He(2^3S)+H_2 Penning sink cannot act in the _adv solve (no n_{H_2} there), so the _adv outputs, and the transit module that reads them, carry the *equilibrium* molecular-base triplet suppression (39× at the HD 209458 b base, 430× on the hot-Uranus case) without an advection correction. This is acceptable where _adv is physically meaningful (the atomic/ionized escape flow above the $H_2 \rightarrow$H front); below the front the atomic _adv approximation is outside its validity domain anyway. A molecular-aware _adv is a separate extension, naturally paired with the open "molecular advection" item in the Tier-2 remaining list.

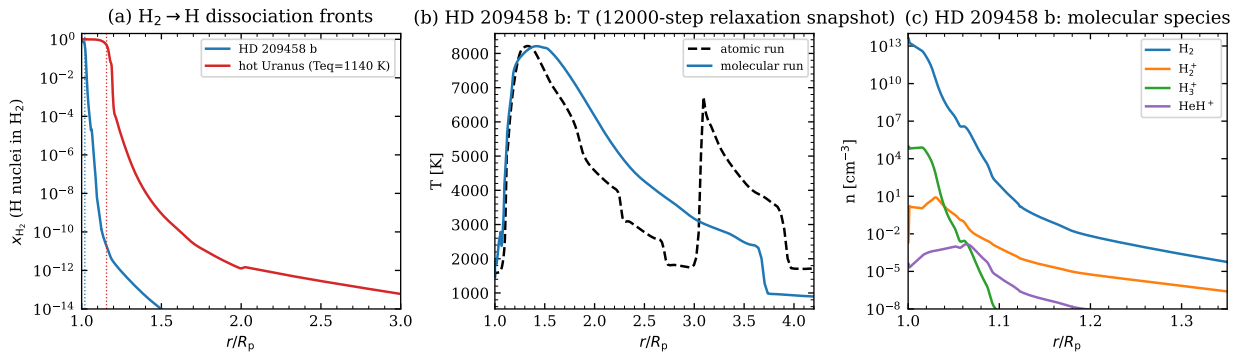


Figure 1: Tier-2 molecular gates (12000-step relaxation snapshots). (a) The H nuclei fraction bound in H_2: on HD 209458 b the base is fully molecular and a sharp $H_2 \rightarrow$H front sits at $r = 1.020\,R_p$ (dotted); on the cooler, low-gravity hot Uranus the front sits at $1.156\,R_p$. The cell-to-cell flip-flop just above the front marks the chemically bistable zone of *equilibrium* H_2 chemistry: kinetics+advection smooth it (the documented Tier-2 caveat). (b) The molecular run's temperature agrees with the atomic run away from the (still-relaxing) front region. (c) Molecular species on HD 209458 b: H_3^+ (green) and the molecular ions are confined below the front, as expected for a hot Jupiter.

Trace metals in the same system (2026-08-13). The molecular network and the metal ionization stages are solved together in `System_HeH_mol_metals`, so a `metals.inp` may sit in a molecular run. They cannot be solved apart: in the shielded molecular base the metals are the dominant electron donors, and the free electron density is shared by both sets of rows.

Remaining Tier-2 work (open). Molecular *advection* (the local chemical-equilibrium treatment cannot reproduce Koskinen et al.'s advective replenishment of high-altitude H_2 on cool Neptunes); the dissociative/double photoionization channels (P4/P5); the 4.48 eV dissociation energy sink of the *thermal* channels R12/R14; a diatomic γ; a molecule-aware `_adv` post-process; and the GJ 1214 b He-halving validation (Taylor et al. 2026) with a full M-dwarf setup. Lyman–Werner photodissociation (§9.) and the metals+molecules merge left this list on 2026-08-13.

4.4 Tier 3: `base.inp` and `run_lower.py`

The handoff file. An optional `base.inp` in the run directory is read by `input_read` *before* the derived constants, so its overrides propagate consistently (v_0, b_0, grid, normalization):

key	unit	overrides	used by
`T_base`	K	T_0 (base temperature)	thermal BC, EOS scales
`r_base`	R_J	"Planet radius" (= r_0 of 1 μbar)	grid, gravity, areas
`HeH_base`	-	He/H number ratio	composition, He diffusion
`Kzz_base`	cm^2/s	`he_kzz`	`He_diffusion` homopause
`q_H2_base`	-	`q_h2_base` (q_{H_2} at the base)	`Molecular base` particle count
`p_base`	bar	`p_base_bar`	level of the handoff; the q_{H_2} fit
`<El>_H_base`	-	`X_<El>`, hence `melem_ab`	metal system, EOS budget, base BC

Every override is echoed to stdout; an absent file is a strict no-op. Unknown keys and # comments are ignored, so a richer lower-atmosphere model may carry extra metadata in the same file.

Elemental reservoirs in the handoff (2026-08-26). `<El>_H_base` is one key per element, spelled with any of the ten element symbols of `species_table` (`C N O Mg Si Ca Na K S Fe`), e.g. `O_H_base 4.90e-4`; the value is the El/H *nuclei* ratio at the handoff level. It overrides `metals.inp` for that element and turns the metal system on by itself when it is the only nonzero abundance: `thereis_metals`, `melem_ab` and `thereis_lowIP_metal` are all derived after the handoff, so a handoff element behaves exactly like a `metals.inp` one. On a restart it also renormalizes that element's whole loaded column onto the stated ratio, preserving the ionization split and the radial shape (`load_IC.f90`). Every key of `base.inp` now carries a *category* (provenance, EOS boundary, elemental reservoir, initial guess, boundary constraint), which is what states what the value is allowed to do to the wind; the full table is `md/input_schema.md` §2c and the user manual's Tier-3 section. Neither `run_lower.py` nor `vulcan_to_base.py` writes `<El>_H_base`: VULCAN's networks are H/C/N/O(/S), so the only element ratios it could hand over are its own inputs, and the atomic-metal release is outside its scope.

Tier 3 as a profile, not a level (2026-08-27). The scalars above state the lower atmosphere at one level. The key `Lower atmosphere profile: <file>` states it over an *interval* of pressure instead: the lower model's own solution, tabulated deep-to-shallow and reaching above the matching level, handed over as one table. The reader is `src/modules/files_IO/lower_atmosphere_profile.f90` and the schema module `src/utils/lower_profile_schema.py`. Two things ask for the interval. The first is eddy mixing: the homopause is where K_{zz} crosses the molecular diffusion coefficient, which a single `He_Kzz` cannot locate, so the profile supplies K_{zz} as a column interpolated

onto the grid. The second is the elemental-flux closure of §6.3. At its matching pressure the profile sets T_0, R_0, $p_{\rm base}$, $q_{\rm H_2}$, He/H and each elemental reservoir it carries; `n_tot` and `rho` are carried but not imposed. Because it is then the single source of those quantities, a `base.inp` beside it may no longer state them: the EOS-boundary, elemental-reservoir and boundary-constraint keys are refused by name, only provenance comments and diagnostic keys remain allowed, and such a file must carry a `# solution_id` matching the profile's or the run stops. Two producers write the format: `src/utils/photochem_to_lower_profile.py` (the production path, with the temperature structure either prescribed or solved by its radiative–convective climate model) and `src/utils/vulcan_to_lower_profile.py` (the cross-check path, VULCAN carries no climate model). Column-by-column schema: `md/input_schema.md` §2d; example: `examples/17_lower_profile/`.

The driver. `src/utils/run_lower.py <run_dir> --r1bar <R_J> [--guillot] [--tint T] [--kzz K]` reads $M_{\rm p}$, $T_{\rm eq}$, He/H from the run directory's `input.inp`, integrates the same column as §4.1 in Python, and writes `base.inp`. Two temperature models: isothermal $T_{\rm eq}$ (default; identical to the Fortran Tier-1 column) and the Guillot (2010) semi-grey global-average profile

$$T^4(\tau) = \frac{3T_{\rm int}^4}{4}\left(\frac{2}{3}+\tau\right) + \frac{3T_{\rm eq}^4}{4}\left(\frac{2}{3}+\frac{1}{\gamma\sqrt{3}}+\left[\frac{\gamma}{\sqrt{3}}-\frac{1}{\gamma\sqrt{3}}\right]e^{-\gamma\tau\sqrt{3}}\right), \qquad \tau = \kappa_{\rm IR}\,p/g, \tag{6}$$

with standard hot-Jupiter defaults ($\kappa_{\rm IR} = 10^{-2}\,{\rm cm^2 g^{-1}}$, $\gamma = 0.4$, $T_{\rm int} = 100\,{\rm K}$). The file records the base $q_{\rm H_2}/q_{\rm H}/q_{\rm He}$ as comments and prints a molecular-base warning when $q_{\rm H_2} > 0.3$. The upgrade path (running `VULCAN` for the composition and a real RC model for $T(p)$) writes the *same* `base.inp` keys, so nothing on the EXHALE side changes.

VULCAN upgrade implemented and demonstrated (2026-07-03). The public `VULCAN` code (cloned to `EXHALE/VULCAN/`; FastChem compiled) was run to steady state for HD 189733 b (SNCHO-2025 photochemical network, 2356 steps) and converted with the new `src/utils/vulcan_to_base.py` (.vul → `base.inp`: photochemical $q_{\rm H_2}/q_{\rm H}/q_{\rm He}$ at 1 μbar, hypsometric $r_{\rm base}$ with VULCAN's own $\mu(p)$ and $T(p)$, molecular mixing ratios as comments). **Result:** at 1 μbar the photochemical base is $q_{\rm H_2} = 0.63$, $q_{\rm H} = 0.23$, *photochemistry dissociates ∼11× more H than the chemical-equilibrium column* ($q_{\rm H} = 0.020$), directly quantifying the Tier-1 caveat (Moses 2011; Koskinen 2013a); $r_{\rm base} = 1.168$ vs the analytic $1.174\,R_{\rm J}$; $T_{\rm base} = 863\,{\rm K}$ (the Moses11 $T(p)$, cooler than the isothermal $T_{\rm eq} = 1183\,{\rm K}$). The full chain VULCAN → converter → `base.inp` → EXHALE startup override was exercised end-to-end. Caveat: VULCAN's networks are H/C/N/O(/S), the atomic-metal release (Na/Mg/Ca/Fe) has no public analogue and remains with the Lavvas collaboration path; `metals.inp` stays user-supplied. Figure 2 shows the converged composition against the equilibrium H_2 track.

The photochemical $q_{\rm H_2}$ now reaches the solve (2026-08-10). Until then the converter recorded $q_{\rm H_2}$ as a comment while the `Molecular base` correction took its own $q_{\rm H_2}$ from the chemical-equilibrium fit: the very approximation the pre-step exists to replace. `base.inp` now carries two further read keys, `q_H2_base` (the photochemical H_2 volume mixing ratio at the base) and `p_base` (the level it refers to, at which the fit is evaluated so that fit and handoff always describe the same level). Both converters write them, and the base particle count is built in one place (`comp_ntot_bc` in `composition.f90`) from the photochemical value when it is present and from the fit otherwise, so a `base.inp` without the key behaves exactly as before. For HD 189733 b the VULCAN value $q_{\rm H_2} = 0.6181$ against the fit's 0.8384 moves

the subtraction from `ntot_bc` by 0.074 per nucleus (∼7%), which the base density inherits in `Base BC: pressure` mode.

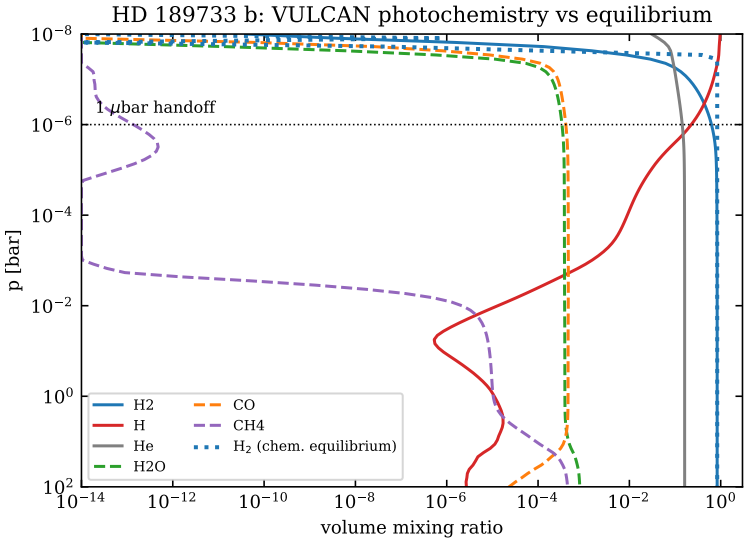


Figure 2: HD 189733 b lower/middle atmosphere from the converged VULCAN run (solid/dashed: H_2, H, He, H_2O, CO, CH_4) compared with the chemical-equilibrium H_2 mixing ratio evaluated along the same $T(p)$ (dotted). Above $\sim 10^{-4}$ bar photochemistry dissociates H_2 far beyond equilibrium; at the 1 μbar handoff (horizontal dotted line) q_H = 0.23 versus the equilibrium 0.020: the $\sim 11\times$ underestimate quantified.

5. Example applications: HD and WASP planets

The approach was applied to the four planets modeled with EXHALE (`examples/13_lower_atmosphere/`; R_{1bar} = broadband transit radius; T_{eq}, M_p, He/H from each run's `input.inp`).

planet	M_p [M_J]	T_{eq} [K]	He/H	R_{1bar} [R_J]	input R_0 [R_J]	r_0(1 μbar) [R_J] chem.eq.	atomic	r_0^{Gui} [R_J]	q_{H_2} (base)	μ_{base} [amu]
HD 209458 b	0.720	1450	0.0833	1.36	1.401	1.472	1.582	1.466	0.831	2.25
HD 189733 b	1.237	1183	0.0833	1.138	1.193	1.174	1.205	1.171	0.838	2.26
WASP-121 b	1.182	2358	0.0851	1.753	2.2075	2.010	2.133	1.966	0.005	1.24
WASP-52 b	0.460	1304	0.0204	1.27	1.270	1.437	1.615	1.428	0.837	1.95

Isothermal-T_{eq} column (chem. equilibrium + fully-atomic bracket) and the Guillot variant r_0^{Gui} (T_{base}^{Gui} = 1313, 1072, 2136, 1181 K respectively). "input R_0" is the base radius currently used in the production runs.

Interpretation (per planet):

- **HD 189733 b: consistent.** The production base radius (1.193 R_J) falls *inside* the derived bracket [1.174, 1.205]: the adopted setup is validated by the column, not merely assumed.

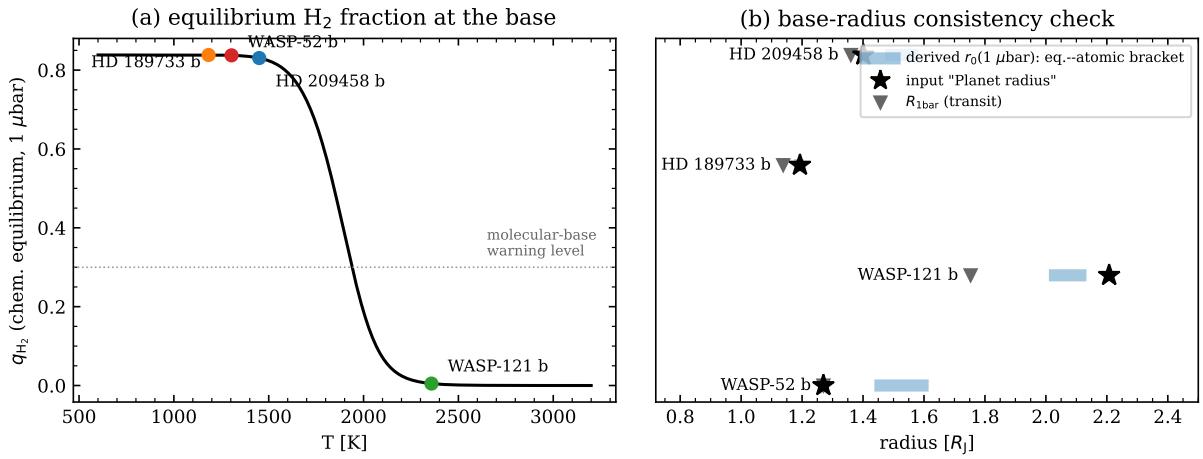


Figure 3: Tier-1 analytic column applied to the four planets. (a) The chemical-equilibrium H_2 mixing ratio at 1 μbar (Eq. 2) as a function of temperature: the three $T_{eq} \lesssim 1500$ K planets sit on the fully molecular plateau, while WASP-121 b ($T_{eq} = 2358$ K) is atomic even in equilibrium. (b) Derived 1-μbar base radius (equilibrium–atomic bracket, shaded) versus the production "Planet radius" (star) and the transit radius (triangle): HD 189733 b falls inside its bracket, HD 209458 b sits ∼5% below, WASP-121 b lies just above (dayside-hot lower atmosphere), and WASP-52 b's transit-radius shortcut puts the base 0.17–0.35 R_J too deep.

- **HD 209458 b: input ∼5% low.** The production value 1.401 R_J sits just below the bracket [1.472, 1.582]. Given the isothermal-T_{eq} assumption this is within the method's spread, but it suggests the 1-μbar level may lie slightly higher than assumed; the effect on $\dot{M}$ is bounded by the Salz base-sensitivity results ($\lesssim$ 10%-level).

- **WASP-121 b: the atomic base is *verified*.** At $T_{eq} = 2358$ K the equilibrium base is already atomic ($q_{H_2} = 0.005$, $q_H = 0.92$, $\mu = 1.24$): for this ultra-hot Jupiter EXHALE's atomic assumption holds *even in equilibrium*, with no appeal to photochemistry. The production base radius 2.2075 R_J (derived earlier from the Huang et al. 2023 Table-3 $\log g$ anchoring) lies ∼4–10% above the isothermal-T_{eq} bracket [2.010, 2.133]: consistent with a dayside lower atmosphere hotter than the redistribution-averaged T_{eq}, which extends the column. Note the deep layers are still molecular (q_{H_2}(1 bar) ≈ 0.83), so μ varies from 2.25 to 1.24 along this column: the equilibrium track is *shorter* than the atomic bracket here because of its heavier deep-layer μ.

- **WASP-52 b: misplaced base, impact quantified.** The production runs use the transit radius itself (1.27 R_J) as the 1-μbar base, the common shortcut, but the column places that level at 1.44–1.62 R_J, i.e. the base is 0.17–0.35 R_J (13–27%) too deep. *Recomputation* (the `fxuv1p0_he98` configuration rerun Newton-converged with the corrected equilibrium-column base $r_0 = 1.437\,R_J$, all else identical): $\log \dot{M}$ 11.97 → 12.14 (×1.5), but the He 10830 observable barely moves, line-center absorption 27.11 → 27.56% (×1.02) and equivalent width 0.713 → 0.722 Å (×1.01). The line-forming effective radius is set by the τ=1 surface of the extended wind, which stays nearly fixed in *physical* units (the larger, less-bound base raises $\dot{M}$, but the He 2^3S column at a given physical radius is almost unchanged: peak n_{2^3S} 150.5→147.6 cm^{-3}). Conclusion: the base misplacement biases the *mass-loss rate* at the ×1.5 level but appears not to bias the He 10830 scan conclusions materially. Figure 4 shows the two line

profiles. Also note this run family's low He/H (= 0.0204, the "He98" abundance scan), visible in the light $\mu_{\rm base}$.

- **Common pattern.** All three $T_{\rm eq} \lesssim 1500$ K planets have strongly molecular *equilibrium* bases ($q_{\rm H_2} \approx 0.83$, the fully-molecular limit of the fit): quantifying that their atomic bases rest on *photochemical* dissociation (Moses et al. 2011; Koskinen et al. 2013a) rather than thermal, which is precisely what the Tier-3 `VULCAN` upgrade would supply from first principles. The Guillot column is systematically slightly more compact (cooler skin above the photosphere): a $\lesssim$0.5% radius effect at fixed assumptions.

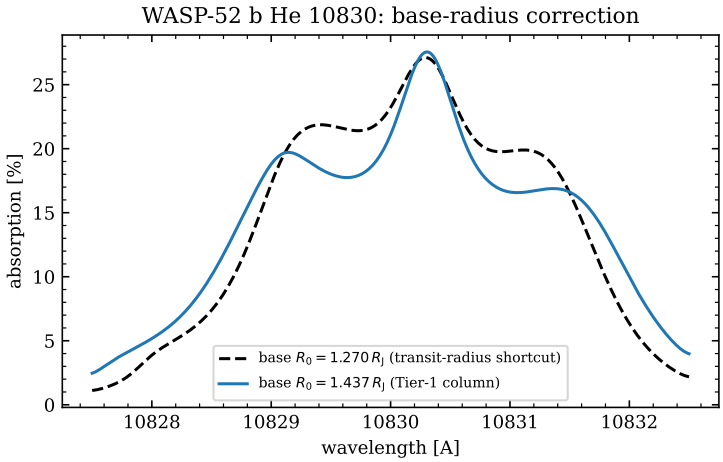


Figure 4: WASP-52 b He 10830 transmission (instrument+rotation convolved) for the old transit-radius-shortcut base ($R_0 = 1.270\,R_{\rm J}$, dashed) and the corrected Tier-1 column base ($R_0 = 1.437\,R_{\rm J}$, solid), both Newton-converged with the same binary. The line core (27.11 vs. 27.55%) and equivalent width (0.713 vs. 0.722 Å) are essentially unchanged (the τ=1 line-forming surface of the extended wind is fixed in physical units) with only a mild reshaping of the shoulders; the base misplacement instead biases $\dot{M}$ ($\log \dot{M}$ 11.97→12.14, ×1.5).

6. Using and connecting the results

6.1 Recommended workflow for each planet

1. **Column check (always, free).** Add `Lower column: <transit radius>` to `input.inp` (or run `run_lower.py`). Read off: does the current "Planet radius" fall inside the [chem.eq., atomic] bracket? Is $q_{\rm H_2}$(base) small?

2. **Decide the tier.** $q_{\rm H_2} \lesssim 0.1$ (e.g. WASP-121 b): the atomic model is verified, proceed as usual. $q_{\rm H_2} \sim 0.8$ with $T_{\rm eq} \sim$ 1000–2000 K (typical hot Jupiters): the atomic base rests on photochemistry, proceed, but note the assumption; a genuinely cool planet (warm Neptune / sub-Neptune) needs Tier-2 physics before its He 10830 predictions are trusted.

3. **Adopt the geometry.** If the input radius lies outside the bracket (WASP-52 b), either correct "Planet radius" by hand or copy the driver's `base.inp` into the run directory so EXHALE adopts r_0, T_0, He/H and K_{zz} automatically (echoed at startup).

4. **Run EXHALE** with the usual two-stage convergence (+ Newton finish for quantitative $\dot{M}$), then `EXHALE_transit.py` for the transit spectra.

5. **(Optional) iterate** once a real lower-atmosphere model is attached (§6.3).

6.2 Where each quantity connects

quantity	produced by	consumed by / effect
$r_0(1\,\mu\mathrm{bar})$	Tier 1 / `base.inp`	grid inner radius, gravity b_0, irradiated area; TPM continuum radius
bracket $[r_0^{\mathrm{eq}}, r_0^{\mathrm{atom}}]$	Tier 1	consistency check on the input radius; systematic-error bar on r_0
T_{base}	driver (T_{eq} or Guillot)	thermal BC T_0; v_0, p_0, b_0 normalization
$q_{\mathrm{H_2}}$(base)	Eq. (2)	tier-decision switch; `Molecular base` correction; Tier-2 initial condition
He/H (base)	`base.inp`	composition; the `He_diffusion` reservoir value
K_{zz}	`base.inp`	`he_kzz`: homopause of the He/H (and metal) diffusive separation
$\Lambda_{\mathrm{H_3^+}}(T, n_{\mathrm{H_3^+}}, n_{\mathrm{H_2}})$	`h3p_cooling`	(Tier 2) new term in the `cool` array of the energy equation
R1–R23 rates	`mol_rates`	(Tier 2) source terms of `System_HeH_mol`; $\mathrm{H_3^+}$/$\mathrm{HeH^+}$ densities for the cooling and for IR-diagnostic post-processing
metal release n_X/n_{H}	(Tier 3, VULCAN)	`metals.inp` values, replaces assumed solar totals for the Mg II/Ca II/Na I transmission lines

6.3 Feedback loop

One-way coupling (lower → `base.inp` → EXHALE, with the Taylor-style tolerated composition discontinuity) is the default.

The elemental-flux closure is implemented (2026-08-27). The two-way iteration runs on the elemental fluxes, driven by `src/utils/element_flux_closure.py`. The chemistry is told a trial elemental flux through its upper boundary (`--trial-flux-H`/`--trial-flux-He` of `photochem_to_lower_profile.py`, $\mathrm{g\,s^{-1}}$ outward positive: the Lavvas et al. 2014 prescription in flux form), the resulting profile becomes the wind's base state, and the wind's own elemental flux is measured face by face by `write_element_flux_profile` (`binary_element_diffusion.f90`),

$$F_{\mathrm{He}} = 4\pi r^2(\rho X v + J), \qquad F_{\mathrm{H}} = 4\pi r^2(\rho(1-X)v - J),$$

and reduced to one median and one relative spread per element over a radial window. F_{H} is the mass flux of component 1 of the binary split, the hydrogen carriers together with the trace metals and the C and O bound in molecules, so $F_{\mathrm{H}} + F_{\mathrm{He}} = \dot{M}$; the chemistry divides `--trial-flux-H` by m_{H}, so the closure hands it the hydrogen-nucleus part $F_{\mathrm{H,nuc}} = s_{\mathrm{H}} F_{\mathrm{H}}$, with $s_{\mathrm{H}} = m_{\mathrm{H}} n_{\mathrm{H}}/m_1$ the hydrogen-nucleus share of component 1's mass in the upwind cell of each face (`lower_profile_F_Hnuc_*` and `steady_F_Hnuc_*` in `EXHALE_resolved.out`; update log stage 3 section 134: before 2026-10-10 the metal mass was counted as hydrogen, 0.44 percent on LHS 1140 b, and $F_{\mathrm{H,nuc}}/F_{\mathrm{H}} = 0.988$ on the `lower_profile` case with solar metals). The solution is the fixed point $\Phi_{\mathrm{El}} = F_{\mathrm{El}}(\Phi_{\mathrm{El}})$, reached by an under-relaxed update $\Phi^{(k+1)} = \Phi^{(k)} + \omega(F^{(k)} - \Phi^{(k)})$ with $\omega = 0.5$ halved (floor 0.125) whenever the residual fails to fall. The convergence test reads the *undamped* residual, so a small ω cannot buy a false convergence, and a residual smaller than the spread

of the window it was measured on is reported `UNRESOLVED` rather than converged. The driver reads the overlap window first and falls back on the steady window $r \geq r_{\rm esc}$ when the overlap is unmeasurable or too ragged, recording which window supplied the number.

Still one-way. The radiation half of the loop is not implemented. EXHALE already computes the attenuated stellar spectrum on its energy grid, so the *transmitted XUV flux at the base*, $F_\nu(r_0) = F_\nu e^{-\tau_\nu(r_0)}$, could be dumped and handed to the photochemical model as its top irradiation; today the adapters take a stellar flux file and a dilution instead. Taylor et al. found $\sim$3 iterations sufficient for the analogous Lyα loop; the same is expected here since the wind is insensitive to base details at the $\lesssim$10% level (Salz et al. 2016).

6.4 The element operator: what it conserves, and where its boundary comes from

The closure above rests on the same operator that carries the He/H and metal separation through the wind, `binary_element_diffusion.f90`. Three statements about it were settled on 2026-09-10 and belong beside the flux numbers, because two of them change what those numbers mean.

The projection now conserves the mixture mass. After each element update the species vector is projected back onto the new element totals by `project_elements`. It used to take the mixture mass as $m_{\rm sum} = m_1 n_{\rm H} + m_{\rm He} n_{\rm He}$ with m_1 the *reservoir* mass per hydrogen nucleus, i.e. the trace metals at their `metals.inp` abundances, while the cell carries its metals at whatever the trace transport left it. The difference was written into hydrogen at every call. MEASURED at one cell of the certification column whose metal/H was 0.024401 against the reservoir's 0.028227, $\sum_i m_i f_i$ went $1.0000000 \rightarrow 1.0000636$ in a single step: mass created. A second leak of the same kind: the trace-metal transport wrote metal densities with no counter-flux in the hydrogen they diffuse through (3.0×10^{-3} on the suite column with the metals frozen).

`mixture_mass_split` now reads the mass the cell's species actually carry, component 1 is scaled as a whole, and one projection after the metal loop puts the metals' mass change back onto hydrogen. Mass closure of the certification column after the relaxation went $3.37 \times 10^{-3} \rightarrow 2.3 \times 10^{-15}$, and of the handed-back atomic candidate $6.44 \times 10^{-3} \rightarrow 1.94 \times 10^{-9}$, the file's own precision. A run with metals off is unaffected by construction ($m_1 = m_{\rm H} = 1$ exactly). The counter-flux is closed on hydrogen but not on momentum or internal energy; that is a trace approximation and is named as one at the routine.

This is not a cosmetic repair for anyone using the coupling: the `lower_profile` regression case, the one that hands the lower atmosphere over as a profile, had a stored reference 3.6×10^{-4} out of closure (O/H 4.61×10^{-4} against the reservoir's 4.90×10^{-4}). A 1500-step cold start goes $6.77 \times 10^{-3} \rightarrow 3.14 \times 10^{-9}$ in closure, with T at $1.6\,R_{\rm p}$ moving 1.3% and H I 6.3%. Its reference was re-snapshotted at the 2026-09-10 gate.

The boundary of the residual is the caller's, deliberately. The two relaxation solvers (`element_diffusion_step`, `solve_trace_element_in_hydrogen`) refill the upper ghosts of their own working arrays before every pass, while `element_transport_residual`, which measures the same balance on a state, reads the ghosts the caller left. Making the residual pose its own zero-gradient top was tried and reverted. It is physically the right condition and it changes the

residual by one element's outermost row in its twelfth digit and nothing else (the production callers already hand the operator ghosts equal to the cell to one to four units in the last place), but the atomic element rows of the stationary solve do not survive a last-bit change of those ghosts, whatever makes it: a control that merely multiplies the same three ghost quantities by $1+2\times10^{-16}$ and poses no boundary at all moves them just as far. The amplifier is the Krylov leg at its one-digit tolerance, which crosses that tolerance on a different vector and sends the acceptance test down a different branch a few iterations later. The asymmetry therefore stays in the tree, documented at the routine, which records the size of the caller dependence (1.19×10^{-2} of the row's scale on a stale caller) rather than assuming it away.

The open item: the layer does not conserve the element flux. The window reduction of the flux closure above is also the honest statement of how well the operator conserves what it transports. MEASURED on the atomic candidate state, as the relative radial spread of $F_{\rm El}$ about its median:

window	$F_{\rm He}$ spread	$F_{\rm H}$ spread	face mass-flux spread
layer, $r<1.10$	9.79	9.09	9.25
wind, $r\geq1.20$	2.71×10^{-2}	1.36×10^{-2}	3.99×10^{-3}
escape, $r\geq2.0$	2.01×10^{-3}	2.14×10^{-3}	1.55×10^{-3}

The mass-flux spread the same state is usually quoted with, 9.8×10^{-7}, is a *cell-centered* number and says nothing about these: the element fluxes are three to four decades worse in the same state. The projection fix buys a factor four in the layer ($39.3\rightarrow9.79$ for helium) and nothing in the wind, which is what a mass-closure repair should do, since the leak was largest where the composition changes most. A spread of 9.8 means the layer's element fluxes are not conserved at all, and the standing base wave is the suspect. The consequence has been taken: the certification gates the element and carrier rows only at $r\geq1.20$, and the layer's rows are reported and do not gate until this number is repaired; `certification.f90` states that reason at the declaration of its two regime radii. Numbers here are READ from the certification tolerance anchoring measurement, not re-measured here.

6.5 Consistency cautions

- Do not double-count composition: if `base.inp` sets He/H from a lower model, the `metals.inp` abundances should be the same model's *atomic release* values, not solar totals.
- `base.inp` overrides win over `input.inp` for T_0, R_0, He/H (echoed); remove the file to return to the manual values.
- The Tier-1 column is isothermal and spherical: near-RLO planets (WASP-121 b) need the Roche correction before the bracket is taken quantitatively, and dayside-hot lower atmospheres extend beyond the $T_{\rm eq}$ column (both visible in the WASP-121 b example above).
- The `Molecular base` key changes only the EOS budget; do not combine it with quantitative He 10830 claims for genuinely cool planets: that is Tier-2 physics.

7. Converged Tier-2 solutions (2026-08-13)

Every Tier-2 number quoted above comes from a 12000-step relaxation snapshot: until 2026-08-13 no molecular configuration in the tree had ever reached a steady state (`examples/15_molecular` ran 135694 steps to $du = 2.9 \times 10^{-2}$). The first Newton-converged molecular solutions were obtained on the hot-Uranus gate (`lower_atmosphere_figs/data_g2/input.inp`, the configuration the `mol_base_handoff` regression case runs, $q_{\mathrm{H_2}}^{\mathrm{base}} = 0.75$) once the H_3^+ infrared cooling was moved inside `eval_cool` (`docs/Update_EXHALE_stage1.pdf` §54), so that the marching temperature update and the steady residual balance the same cooling function.

Recipe. Three keys, and they are the whole change:

```
Solver: Newton 5.0e-2   Resid tol: 1.0e-5   Max steps: 150000
```

`Solver: Newton 5.0e-2` raises the hand-off threshold, because the du descent of a molecular run is *not monotonic* (on the hot-Uranus gate it reaches 8.8×10^{-2} near step 11500 and bounces back to 7×10^{-1} before resuming), so a run can spend its whole step budget above the 10^{-2} default. `Max steps: 150000` covers the marching warm-up. Marching alone can *never* reach the molecular layer: its H_3^+ cooling time is 1.3×10^7 s at $r = 1.02$ against a CFL-limited step of 0.30 s, i.e. 4×10^7 steps per cooling time. It is the Newton solve that reaches it.

The default residual target is too loose for a molecular run. $\|R\|$ is the maximum relative residual over the wind, and in a molecular run it is set by the first two or three cells above the base, where the energy residual is $\sim 10^3$ times the value anywhere else. Stopping at the default $\|R\| < 10^{-3}$ therefore leaves the molecular layer still cooling: its cell-by-cell energy residual is then $0.7\times$ the local H_3^+ cooling rate, with the same sign in 173 of 180 cells. Re-solving the same state with `Resid tol: 1.0e-5` moves the layer again (by 60–70% in T), and only there does the solution become a fixed point of the procedure: a third solve reproduces T to $\leq 0.14\%$ and the $H_2 \rightarrow$H front to five digits. **A molecular run needs `Resid tol: 1.0e-5`.**

hot-Uranus gate	metals off snapshot → converged	metals on (solar C/N/O/Mg/Ca/Na/Fe) snapshot → converged
residual norm at exit	6.7×10^{-7}	1.4×10^{-6}
$H_2 \rightarrow$H front ($x_{\mathrm{H_2}} = 0.5$)	1.16037 → **1.03581**	1.16244 → **1.07974**
base T (cell 1, pinned by the BC)	1213.42 → 1212.86 K	1213.46 → 1213.45 K
H_3^+ peak [cm^{-3}]	7.03×10^4@1.0419 → $\mathbf{3.75\times10^5}$@1.0148	52.1@1.1355 → $\mathbf{1.01\times10^3}$@1.0789
T at $r = 1.02$	1304.9 → **299.0** K	1312.8 → **928.5** K
T at $r = 1.08$	1302.6 → 1075.5 K	1319.8 → **208.9** K
$\log_{10}\dot{M}$ [g/s], spherical	(not flat) → 10.542	(not flat) → 10.599
mass-flux spread, $r > 1.2$	2.3×10^2–3.8×10^3 → 2.4×10^{-3}	idem

The snapshot $\dot{M}$ is quoted only because the code prints it: its mass flux varies by a factor 2×10^2–4×10^3 across the wind, so no single number represents it.

What the converged solution says physically. The molecular layer collapses to 190–300 K (far below $T_{\rm eq} = 1140$ K), and stops there because the coolant switches itself off: the H_3^+ cooling time at $r = 1.02$ rises from 1.3×10^7 s in the snapshot to 1.3×10^{10} s in the converged state. With metals the collapse is instead accompanied by the saturated ground-term fine-structure lines (C I 609/370 μm at the base, [O I] 63/145 μm above it: 89% and 91% of the local cooling at $r = 1.002$ and 1.040), and H_3^+ never exceeds 2.3%. Either way the layer is cooled by lines treated as *optically thin*, with no thermal-infrared heating from below and no radiative equilibrium: the base temperature is pinned in one cell by the boundary condition and nothing holds the column above it. A real H_2 atmosphere at these column densities is optically thick in exactly these lines and sits near the radiative–convective profile. The converged Tier-2 thermal structure should therefore be read as **the steady state of the model as written, and as a quantitative statement that the model was missing an infrared escape probability below the $H_2 \rightarrow$H front**, not as a prediction of the temperature of a warm Neptune's lower thermosphere. §8. supplies part of what is missing and measures how far it goes. The wind above the front is much less affected: $\dot{M}$ moves 0.085 dex (metals off) and 0.043 dex (metals on) between the 10^{-3} and 10^{-5} solutions, because it is launched above the collapsed layer.

HD 209458 b: the same recipe on a real planet. The second Newton-grade molecular solution is `examples/15_molecular`: the metals-off HD 209458 b configuration of Gate 1, the one that had run 135694 steps without ever converging. The three keys above are the whole change: no warm restart was needed, the run went from the default cold IC to `info = 0` in one pass, 10 min 25 s on 8 threads. PLM $\rightarrow$ WENO3 at step 13299, secondary ionization flipped in at step 23837, JFNK engaged at step 25837 and took $\|R\|$ from 4.031×10^{-2} to $\mathbf{9.542 \times 10^{-6}}$ in 272 iterations. Here too the *du* descent is not monotonic. The residual is dominated by the same place: the worst cell is $j = 1$–3, $r \approx 1.000$–1.001, for 267 of the 272 iterations, split about evenly between the momentum and the energy row.

HD 209458 b, metals off	12000-step snapshot	converged ($\|R\| = 9.5 \times 10^{-6}$)
$H_2 \rightarrow$H front ($x_{H_2} = 0.5$)	1.01975	**1.00891**
base T (cell 1, pinned by the BC)	1466.55 K	1466.36 K
T at $r = 1.005/1.02/1.08$	1909 / 1916 / 2855 K	**401** / 2206 / 7574 K
coldest cell below $r = 1.3$	1463.6 K @ 1.0002	**397.3 K @ 1.0056**
H_3^+ peak	9.57×10^4 cm^{-3} @ 1.0004	$\mathbf{5.18 \times 10^5}$ cm^{-3} @ 1.0017
x_{H_2} at the base	0.9964	0.9963
$\log_{10}\dot{M}$ [g/s], spherical	10.873	10.532
mass-flux spread over $r > 1.05$	1.23	$\mathbf{4.2 \times 10^{-3}}$

The solution is a fixed point of the procedure: re-solving from it with the same tolerance returns `info = 0` at $\|R\| = 9.0 \times 10^{-6}$ and reproduces the front to 1.00879 (0.01%), $\dot{M}$ to 0.002 dex and the H_3^+ peak to 0.4%; only the coldest cell of the collapsed layer still moves (397 $\rightarrow$ 353 K). Note the reload does not preserve the residual: `Load IC? True` re-derives the ionization state from the file, so the second solve starts at $\|R\| = 8.7 \times 10^{-4}$. The physical reading is the hot-Uranus reading, on a hot Jupiter: the molecular layer between the base and the front collapses to $\sim$400 K, H_3^+ carries **100%** of the radiative cooling from the base out to $r \approx 1.005$ (10^4 times the local photoheating at the base), and the same optically thin treatment applies. The front moves inward by 0.011 $R_{\rm p}$ and $\dot{M}$ falls by 0.34 dex relative to the snapshot, so for HD 209458 b the relaxation-snapshot $\dot{M}$ is *not* a substitute for the converged one.

With metals the same planet does not converge. `examples/16_molecular_metals` (the identical configuration plus solar C/N/O) under the same three keys reaches the hand-off (PLM → WENO3 at step 15263, secondary ionization at 61158), and the JFNK then stalls: $\|R\|$ falls from 2.338×10^{-2} to 3.6×10^{-4} over 281 iterations and the line search finds no descent step for 12 consecutive iterations (`info = 2`, worst cell $j = 93$, $r = 1.021$, momentum), after which the run reverts to marching. This is the reverse of the hot-Uranus gate, where the metals-on case converged as readily as the metals-off one, and it is the base-adjacent momentum stall recorded as open item (A), displaced outward to the front. `16_molecular_metals` therefore carries the plain `Solver: Newton` line and has no converged solution.

8. The infrared field of the lower atmosphere (`Base IR field`)

The converged Tier-2 solutions above left one question open: why does the molecular layer sit at 190–300 K, and is the optically thin line cooling the cause? The measurement below says the suspect is the right one for the metals-off case and the *wrong* one for the metals-on case, and the closure follows the measurement.

What the lines actually see. Line-center optical depths of the eight ground-term fine-structure lines, from the converged metals-on solution of the hot-Uranus gate, with the level populations and Doppler widths the cooling module itself uses. $\tau_{\rm up}$ is the column from the cell to the top of the domain (the one the escape probability was built from), $\tau_{\rm dn}$ the column to the bottom.

$r/R_{\rm p}$	T [K]	$\tau_{\rm up}$(C I 609)	$\tau_{\rm dn}$	β	$\tau_{\rm up}$([O I] 63)	$\tau_{\rm dn}$	β
0.9998	1213	0.088	1.7×10^{-4}	0.904	2.01	4.0×10^{-3}	0.211
1.0102	1033	0.069	0.019	0.924	1.57	0.44	0.265
1.0302	820	0.040	0.048	0.955	0.89	1.1	0.421
1.0501	573	0.020	0.068	0.978	0.43	1.6	0.633
1.0749	289	0.005	0.083	0.995	0.093	1.9	0.898
1.0832	188	0.001	0.086	0.998	0.028	2.0	0.968

So **escape was never the problem**: outward the collapsed layer is thin ($\beta \geq 0.90$ for C I, ≥ 0.63 for [O I] above $r = 1.05$), and trapping can only reduce the cooling by a factor of a few at the base. What the treatment left out is the *other* direction. The gas below the base is not vacuum. Taking the base cell opacity and one pressure scale height of an isothermal exponential reservoir (base pressure 9.0 μbar):

line	τ per scale height at the base	pressure where $\tau = 1$	τ at 1 bar
[O I] 63 μm	1.08	17 μbar	1.2×10^5
[O I] 145 μm	0.32	37 μbar	3.6×10^4
C I 370 μm	0.092	106 μbar	1.0×10^4
C I 609 μm	0.046	205 μbar	5.1×10^3
C II 158, N II 205/122, [O I] 44 μm	$\leq 1.2 \times 10^{-10}$	-	$\leq 3 \times 10^{-2}$

Every line that carries the cooling is black downward within one to three scale heights below the model base. The layer therefore faces a blackbody, not a void, and the ratio of the incident mean intensity to the line source function,

$\bar{J}/S = \frac{1}{2}B_\nu(T_{\rm base})/B_\nu(T)$, crosses 1 wherever T drops below ∼610 K: at the coldest cell it is 3.4 (C I 609 μm) to 5.7 ([O I] 63 μm). Those lines were being made to cool a gas that they should have been heating.

But that is not what makes the metals-on layer cold. Two measurements say the fine-structure lines are not the cause of the 190–300 K there.

- **The specific entropy rises monotonically outward through the whole collapsed layer**, p/ρ^γ going from 1.00 at the base to 1.67 at $r = 1.05$ and 5.69 at $r = 1.083$. The gas is net heated everywhere; it is cold because ρ has fallen by a factor 10^2. The layer is an expansion, not a radiative collapse.
- **The metals-off solution collapses further with no fine-structure cooling at all**: 115 K at $r = 1.033$, where its total cooling is 3.8×10^{-12} erg cm^{-3} s^{-1}.

The energy budget makes the same point directly: the metals-on cooling is 36% of the local photoheating at $r = 1.01$ and 2–16% above $r = 1.02$, and heat − cool is balanced by the expansion term $\rho v[\mathrm{d}\varepsilon/\mathrm{d}r + p\,\mathrm{d}(1/\rho)/\mathrm{d}r]$ to within a few percent. The one place radiation dominates is the metals-off base: there H_3^+ carries 100% of the cooling, the entropy *falls* by a factor 3 between the base and $r = 1.02$, and the layer really is radiating itself down.

The closure. `Base IR field: True` (default `False`) gives both families of infrared coolant the field they sit in. The lower atmosphere is taken to be black at their wavelengths and to radiate $B_\nu(T_0)$ over the sky fraction $f = 1 - \sqrt{1-(R_{\rm p}/r)^2}$, i.e. the whole lower hemisphere at the base and the usual dilution far away.

For the eight fine-structure lines the incident field enters *inside* the ground-term statistical equilibrium, as the photon occupation number

$$\bar{n} = \frac{\beta_1(\tau_{\rm dn})\, f}{\exp(E_{ul}/kT_0) - 1}, \tag{7}$$

with radiative rates $\beta A(1+\bar{n})$ down and $\beta A(g_u/g_l)\bar{n}$ up, so the returned power is the *net* one, emission minus absorption. The escape probability becomes two-sided, $\beta = \beta_1(\tau_{\rm up}) + \beta_1(\tau_{\rm dn})$ with β_1 the single-face Hollenbach & McKee (1979) / de Jong et al. (1980) form; with the field off it reduces to $2\beta_1(\tau_{\rm up})$, the previous value, bit for bit. Each line then stops cooling at its own radiative equilibrium temperature, 575.8 K (C I 609 μm) to 641.7 K ([O I] 44 μm) for $T_0 = 1140$ K and half-sky coverage, and heats below it.

For H_3^+ the emission is a total cooling function, not a set of lines resolved in the code, so the absorption is taken from the same function at the radiating temperature, which is what Kirchhoff's law makes of an emission integral: $\Lambda_{\rm net} = n(H_3^+)\,4\pi\, s(T, n_{H_2})\,[W(T) - W_{\rm dil}\,W(T_0)]$, with $W(T)$ the MiZATeP cooling function and $W_{\rm dil}$ the dilution of the field. This is the same net-exchange form the H_2, H_2O and CO channels use (§10.), and it is bounded at every temperature. Its radiative equilibrium temperature is 958.5 K for $T_0 = 1140$ K and $W_{\rm dil} = 1/2$ on the MiZATeP table (header of `h3p_cooling.f90`); with the Miller et al. (2013) fit used until 2026-10-10 it was 975.4 K. Between 2026-08-13 and 2026-08-31 the exchange was instead closed on one effective band, the ν_2 fundamental at 2521.3 cm^{-1} ($E/k = 3627.5$ K), as $\Lambda_{\rm net} = \Lambda_{\rm emit}(T)\,[1 + \bar{n} - \bar{n}\,e^{E/kT}]$, with equilibrium temperatures of 936.1 K and then 941.2 K. Pairing a total emission fit with the Boltzmann factor of one of its transitions diverges below about 150 K and is wrong by a factor 1.5–1.9 over 400–1000 K; §112 of `docs/Update_EXHALE_stage1.pdf` measures it and records what moved. The approximations and where

they break are written at `h3p_net_cooling_rate` (`src/modules/lower_atmosphere/h3p_cooling.f90`) and at `fine_structure_line_transfer` (`src/modules/radiation/Cool_coeff.f90`). Fe II (a precomputed statistical-equilibrium table), the coronal remainders and every permitted line keep the optically thin, no-incident-field limit; the estimated heating the remainders would add is ∼0.4% of the losses of this layer.

What it does. Both runs restart the converged `Resid tol: 1.0e-5` solutions of §7. with the switch on, and both return `info = 0`.

hot-Uranus gate	metals off		metals on	
	field off	**field on**	field off	**field on**
residual norm at exit	6.7×10^{-7}	1.3×10^{-6}	1.4×10^{-6}	9.9×10^{-6}
T at $r = 1.005$	626.8 K	**910.0 K**	1099.0 K	1098.9 K
T at $r = 1.02$	299.6 K	**859.4 K**	928.9 K	928.9 K
T at $r = 1.03$	140.2 K	**840.2 K**	820.0 K	820.1 K
T at $r = 1.05$	476.8 K	**753.0 K**	573.0 K	573.4 K
coldest cell	115.1 K @ 1.0334	**229.7 K @ 1.0947**	187.9 K @ 1.0831	189.1 K @ 1.0831
H_2 →H front	1.03582	**1.08833**	1.07976	1.07979
H_3^+ peak [cm^{-3}]	3.75×10^{5}	3.19×10^{5}	1.01×10^{3}	1.00×10^{3}
$\log_{10}\dot{M}$, spherical	10.539	10.617	10.596	10.596
mass-flux spread, $r > 1.2$	2.4×10^{-3}	2.4×10^{-3}	2.4×10^{-3}	2.4×10^{-3}

Stale, and not re-measurable from here: the field-on columns and the predicted floors below were computed with the single-band H_3^+ closure, which was replaced on 2026-08-31 (§112 of `docs/Update_EXHALE_stage1.pdf`). The converged solutions they restart from no longer exist as run directories, so the table is left as the record of what that closure gave. The same four-way comparison re-measured with the current closure, at the 12000-step protocol of the `mol_ir_bands` regression case, is in §112.

The metals-off layer stops collapsing and settles *on* the H_3^+ radiative equilibrium curve: the predicted floor with the local sky fraction is 911 K at $r = 1.005$, 886 K at 1.02 and 874 K at 1.03, and the solution sits at 910, 859 and 840 K, just below, by the margin the expansion takes out. Its H_3^+ channel becomes a net *heating* term (-1.1×10^{-7} erg cm^{-3} s^{-1} at $r = 1.02$) instead of the -1.6×10^{-10} of cooling it had. The front moves out by $0.053\,R_p$ and $\dot{M}$ by +0.078 dex. It is a fixed point of the procedure like the solutions above. The metals-on layer does **not** move (0.1–1.2 K), for the reason measured above: its temperature is set by the expansion, and the fine-structure lines it does carry are 5–20% of the local heating. They now correctly turn into a heating term above $r \approx 1.06$, which is physically right and numerically almost invisible.

What is still open. The metals-on molecular layer, and the outer part of the metals-off one past the H_2 →H front (229.7 K at $r = 1.095$), remain far below any radiative equilibrium temperature, and this closure cannot reach them: it only gives the *existing* infrared coolants their incident field, and above the front there are no molecular coolants left. Holding that gas requires a continuum infrared coupling to the deep atmosphere (H_2 collision-induced absorption and the H_2O/CH_4/CO bands), which the model does not have: the whole radiative budget between the base and the front is

line channels with a radiative time of 3×10^8 s against a flow time of 5×10^6 s. Open item (G) is therefore narrowed, not closed.

9. H_2 photodissociation in the Lyman–Werner bands (`Stellar LW flux`)

The Tier-2 network inherited from Koskinen et al. (2022, Table 1) has no photodissociation of neutral H_2: its photorates start at the 15.4 eV photoionization edge, so below that the only H_2 losses are thermal (R12), electron impact (R14) and ion chemistry. Their own note says the omission is deliberate and that adding it (Backx et al. 1976 cross section, dissociation probability 0.125) moved their $\dot{M}$ by $\leq 1.4\times$. The consequence for us is recorded: the network wants a base more molecular than any photochemical code gives, and pinning q_{H_2} at the base (Route 1) patches over the missing physics rather than supplying it. This section supplies it.

The band is not in the code's own radiation field. Checked in the source, not assumed. `set_energy_vectors` builds the photon grid from 13.6 eV up; it extends below the H I edge only when the He $2\,^3$S metastable (4.8 eV) or a low-IP metal is active, and then only to that species' threshold, and `read_sed` selects SED rows by the same `e_low`. So for a numerical SED the 11.2–13.6 eV band is simply not read, and for a power-law SED (every Tier-2 gate case) anything below 13.6 eV is the XUV power law extrapolated downward, which has no relation to a star's FUV. Even where the grid does reach into the band, its opacity is continuum photoionization, whereas Lyman–Werner absorption is a forest of saturated lines whose attenuation is nothing like $e^{-\tau_{\rm continuum}}$. **The band flux is therefore a separate user input, and the transfer is done separately:**

`Stellar LW flux [erg/cm2/s]: 343.0` (912–1110 Å, integrated, at the planet)

Default 0 = off. With `Molecular chemistry` off the key is inert and says so.

Rate. Draine & Bertoldi (1996), ApJ 468, 269 (DB96), published version. Their band is 912–1110 Å: 912 Å is the H Lyman edge, and longward of 1110 Å the H_2 absorptions out of $v = 0$ are negligibly weak (their footnote 4). They characterize a field by the photon flux in that band, $F \equiv c n_{\rm phot}$ (their eq. 21), and tabulate it with the unshielded dissociation rate. For the flat-F_λ spectrum ($u_\nu \propto \nu^{-2}$, their eq. 24) at $\chi = 1$: $F = 1.208 \times 10^7$ photons cm^{-2} s^{-1} (Table 1), $\zeta_{\rm pump} = 3.09 \times 10^{-10}$ s^{-1} with $\langle p_{\rm diss} \rangle = 0.135$ (Table 2), hence $\zeta_{\rm diss}(0) = 4.17 \times 10^{-11}$ s^{-1} (Fig. 7 caption). The dissociation rate per band photon is then an effective cross section

$$\sigma_{\rm LW} = \frac{4.17 \times 10^{-11}}{1.208 \times 10^7} = 3.452 \times 10^{-18}\ {\rm cm}^2, \tag{8}$$

and with the mean photon energy of a flat-F_λ band, $\langle h\nu \rangle = 2hc/(912 + 1110\,\text{Å}) = 12.2635$ eV,

$$k_{\rm LW,thin} = 1.757 \times 10^{-7} F_{\rm LW} \quad [{\rm s}^{-1},\ F_{\rm LW} \text{ in erg cm}^{-2}\,{\rm s}^{-1}]. \tag{9}$$

Reproducing DB96's own Table 1 numerically from their eqs. (22)–(24) confirms the flux convention (Habing 1.2220×10^7 against their 1.222×10^7; ν^{-2} 1.2084 against 1.208; Draine 1978 1.2313 against 1.232, all $\times 10^7$) and the closure of the calibration (the formula returns 4.17×10^{-11} s^{-1} for the ν^{-2} field at $\chi = 1$, their value to 0.04%).

The approximation and its range. $\sigma_{\rm LW}$ depends on the shape of the spectrum *within* the band, because the Lyman/Werner lines sample it unevenly. The same arithmetic for the much softer Draine 1978 field ($\zeta_{\rm pump} = 2.78 \times 10^{-10}$, $\langle p_{\rm diss} \rangle = 0.119$, $F = 1.232 \times 10^{7}$ at $\chi = 1$) gives 2.685×10^{-18} cm^2, 22% lower, and the formula above would overpredict that field's rate by 26%. The two spectra bracket color temperatures 1.3–2.9 $\times 10^4$ K, and DB96 note that PDR properties are insensitive to the spectrum for $T_{\rm color} \gtrsim 10^4$ K. **Take the adopted value as good to $\pm$25% for a stellar FUV band that is not strongly tilted**; a band dominated by a single emission line at one end is outside it.

Self-shielding. DB96 eq. (37), their fit to the full multiline calculation *including line overlap*,

$$f_{\rm shield}(N_{\rm H_2}) = \frac{0.965}{(1+x/b_5)^2} + \frac{0.035}{(1+x)^{1/2}} \exp\left[-8.5\times10^{-4}(1+x)^{1/2}\right], \qquad x = \frac{N_{\rm H_2}}{5\times10^{14}\,{\rm cm^{-2}}}, \tag{10}$$

with $b_5 = b/10^5$ cm s^{-1}, and their eq. (40), $\zeta_{\rm diss} = f_{\rm shield}\, e^{-\tau_{\rm dust}}\, \zeta_{\rm diss}(0)$. It reproduces the exact self-shielding function over $10^{14} < N_{\rm H_2} < 3 \times 10^{21}$ cm^{-2} (their Figs. 1–5).

The rate no longer uses that $f_{\rm shield}$. DB96's fit is calibrated for cold gas in which only the lowest rotational states of H_2 are populated; Richings, Schaye & Oppenheimer (2014), MNRAS 442, 2780, compared it against a level-resolved CLOUDY calculation and found that it overestimates the shielding factor by a factor $\sim$3. Our molecular layer runs at 945–1307 K, where the overestimate measured with our own columns is 2.8–5.4$\times$. The photodissociation rate therefore uses their temperature-dependent eqs. (3.12)–(3.15),

$$S_{\rm self}(N_{\rm H_2}, T) = \frac{1-w(T)}{(1+x/b_5)^{\alpha(T)}}\, e^{-5\times10^{-7}(1+x)} + \frac{w(T)}{(1+x)^{1/2}}\, e^{-8.5\times10^{-4}(1+x)^{1/2}}, \qquad x = \frac{N_{\rm H_2}}{N_{\rm crit}(T)}, \tag{11}$$

with $w(T)$, $\alpha(T)$ and $N_{\rm crit}(T)$ their eqs. (3.13)–(3.15) and the same thermal Doppler parameter: their fit was made for purely thermal broadening, and their Fig. B1 caption states that the agreement with CLOUDY is poorer once turbulence is included, so no turbulent term is added. DB96 eq. (37) is kept, and is still used, for the *band equivalent width* of the shared 912–1110 Å beam: their eq. (39) is the closed-form integral of eq. (37) and of nothing else, and Richings publishes no corresponding equivalent width. `lyman_werner.f90` §2 sets out why the two coexist and which quantity each owns.

b is the H_2 Doppler parameter, taken thermal, $b = (2kT/m_{\rm H_2})^{1/2}$: 3.07 km s^{-1} at 1140 K, next to the 3 km s^{-1} of DB96's own figures. $N_{\rm H_2}$ is the star-ward column, built by the same `calc_column_dens_one` radial integration and `opa_pf` weighting as every other absorber column, from the incoming (pre-solve) H_2 density.

Three things are deliberately *not* attenuating the band, all noted at the code: **dust** (EXHALE's metals are atomic and trace, so there are no grains and the $e^{-\tau_{\rm dust}}$ of eq. 40 is identically 1); the **trace-metal continuum** (the neutral low-IP metals do photoionize inside the band, but at solar abundance and $\sigma \sim 10^{-18}$ cm^2 their optical depth is $\sim 4 \times 10^{-23} N_{\rm H}$, i.e. $\lesssim 0.05$ at a base column where $f_{\rm shield}$ is already below 10^{-4}); and the **H Lyman-series lines** (Lyβ 1025.7, Lyγ 972.5, … all lie in the band), which DB96 include in the equivalent width their fit was built on and state, §4.2, "We will see below that absorption by the H Lyman lines has only a small effect on the H_2 pumping rates". The last is the least controlled of the three, because a wind has a much larger $N_{\rm HI}/N_{\rm H_2}$ than the PDRs DB96 fitted; it can only reduce the rate.

Chemistry and heating. $H_2 + h\nu \rightarrow H + H$ enters the H_2 balance row of the coupled system next to the H_2 photoionization (`mol_heh_rows`, `System_HeH_mol`), so it is solved together with everything else rather than in a side loop.

Both products are neutral H, which the H-nucleus closure supplies automatically; no other row changes. The fragments carry kinetic energy: Black & Dalgarno (1977), ApJS 34, 405, p. 418, "Fluorescent dissociation of H2 gives rise to a pair of energetic hydrogen atoms … for a typical ultraviolet radiation field, the yield is about 0.4 eV per atom pair, but it varies slightly with depth." We adopt 0.4 eV per dissociation, held constant with depth, added to `heat` in `ioniz_eq` next to the Penning heating terms. **The 4.48 eV bond energy is paid by the absorbed photon, not by the gas, and is not a sink of this channel** (a thermal dissociation-energy sink for R12/R14 remains a separate open item). `output/Lyman_Werner.txt` is written whenever a molecular run carries a band flux: r, T, x_{H_2}, n_{H_2}, N_{H_2}, f_{shield}, k_{LW} and the photodissociation heating.

The band flux for a planet. The code's own SED cannot supply it, so it is integrated externally. The VULCAN stellar spectra in the tree (`VULCAN/atm/stellar_flux/`, flux at the *stellar surface*, erg cm^{-2} s^{-1} nm^{-1}) integrated over 91.2–111.0 nm and diluted by $(R_\star/a)^2$:

star / planet	surface band flux	$R_\star$, a	F_{LW} at the planet
Sun, `Gueymard_solar.txt`, at 1 au	2.74×10^4	1 $R_\odot$, 1 au	0.592
HD 209458 b (solar spectrum as proxy)	2.74×10^4	1.155 $R_\odot$, 0.048 au	**343**
HD 189733 b, `sflux-HD189_Moses11.txt`	4.23×10^4	0.805 $R_\odot$, 0.03142 au	600
HD 189733 b, `sflux-HD189_B2020.txt`	1.29×10^5	0.805 $R_\odot$, 0.03142 au	1.83×10^3

(erg cm^{-2} s^{-1} throughout; the corresponding $k_{LW,thin}$ are 1.04×10^{-7}, 6.03×10^{-5}, 1.05×10^{-4} and 3.21×10^{-4} s^{-1}.) The `VULCAN_run_*/atm/stellar_flux/` directories of the workspace are empty (casualties of the 2026-08-09 loss), so no HD 209458 spectrum exists in the tree and the solar spectrum stands in for it; HD 209458 is a G0 star of comparable activity, but **343 erg cm^{-2} s^{-1} is a proxy, not a measurement**. The calibration check on the same file is the solar Lyα irradiance it returns at 1 au, 6.12 erg cm^{-2} s^{-1} against an observed 6–8. The factor 3 between the two HD 189733 spectra is a fair measure of how uncertain a real stellar FUV band flux is, and it dwarfs the $\pm 25\%$ of the band-shape approximation.

What it does: hot-Uranus gate, A/B on the key alone (metals off). Both runs restart the `mol_base_handoff` configuration ($q_{H_2}^{base} = 0.75$, $p_{base} = 10^{-5}$ bar) with the three convergence keys of §7. and `Base IR field: True`, and differ only in `Stellar LW flux`. Both return `info = 0`.

	LW off	**LW on (343)**
residual norm at exit	8.15×10^{-6}	4.86×10^{-6}
$H_2 \rightarrow$H front ($x_{H_2} = 0.5$)	1.08806	**1.08622**
x_{H_2} at the base	0.99921	0.99883
q_{H_2} at the base	0.8618	0.8612
x_{H_2} at $r = 1.05$	0.9777	0.9629
x_{H_2} at $r = 1.10$	2.54×10^{-2}	$\mathbf{1.51 \times 10^{-4}}$
T at $r = 1.005/1.05/1.10$	901.5 / 752.3 / 255.6 K	886.6 / 751.8 / 249.9 K
$\log_{10}\dot{M}$, spherical	10.6196	10.6248

The self-shielding works as designed and is visible in the dump: $N_{H_2} = 4.32 \times 10^{21}$ cm^{-2} and $f_{shield} = 9.79 \times 10^{-7}$ at the base, 2.85×10^{-4} at the front ($r = 1.087$), 0.467 at $r = 1.10$ and exactly 1 above $r = 1.15$, where k_{LW} recovers the unshielded 6.026×10^{-5} s^{-1} computed above. The band is unattenuated in the wind and suppressed by 10^6 at the base.

Route 2 verdict: the physics was missing, but it does not close the gap. The Route 2 criterion of `base_composition_handoff_plan.md` §5 was whether the network's own base composition moves toward the photochemical partition once the missing dissociation is supplied, without pinning it. **It does not**, and the measurement says why. The base q_{H_2} moves from 0.8618 to 0.8612 against a photochemical target of 0.75: 0.5% of a gap of 0.11. Yet Lyman–Werner *is* the largest single H_2 loss term at the base after the change: 5.90×10^{-11} s^{-1} against 2.52×10^{-11} for H^++H_2 (R10 + R13), 5.56×10^{-12} for He^++H_2 and 3.87×10^{-13} for thermal dissociation. The two facts are consistent because the base partition is a formation–destruction balance in which the three-body reaction R15 (H+H+M) sets $n_{H_2} \propto n_H^2$, so

$$\frac{n_H}{n_{H_2}} \propto \left(\text{total } H_2 \text{ destruction rate}\right)^{1/2}, \tag{12}$$

and the square root is brutal: the measured 2.3× rise in the destruction rate raises the atomic fraction by only 1.5×, from 7.9×10^{-4} to 1.2×10^{-3}. Reaching $q_{H_2} = 0.75$ means $x_{H_2} = 0.925$, i.e. an atomic fraction 64× larger, i.e. a destruction rate **4100× larger**: about 3.7×10^{-7} s^{-1}. The unshielded band would supply that many times over (6.0×10^{-5} s^{-1}); it is stopped by the column. The base of this planet sits under $N_{H_2} = 4.3 \times 10^{21}$ cm^{-2}, where $f_{shield} = 9.8 \times 10^{-7}$, and f_{shield} reaches the required $\sim 6 \times 10^{-3}$ only near $N_{H_2} \sim 10^{17}$–10^{18} cm^{-2}, four orders of magnitude shallower.

So Lyman–Werner photodissociation cannot be the reason photochemical codes find more atomic H at 1 μbar. That reading is not new physics: §2 of this memo already records that photochemistry (OH-catalyzed H_2 destruction) dissociates H_2 even at 500–1000 K where thermal equilibrium would not. The catalytic cycles run on O, OH and H_2O, and EXHALE's molecular network is $H_2/H_2^+/H_3^+/HeH^+$ with atomic metals; it has nowhere to put them. Closing the base gap needs those species, not a stronger radiation field. Route 1 (pinning `q_H2_base`) therefore remains the only way EXHALE can carry the photochemical base partition today, and it is now a documented modeling choice rather than a patch over a missing term.

What Lyman–Werner *does* change is the top of the molecular layer, where the column has fallen enough for the band to bite: at $r = 1.10$ the H_2 fraction drops by a factor 168, the front moves inward by $0.0018\,R_p$, and $\dot{M}$ rises by 0.005 dex. Small, and in the direction and of the order Koskinen et al. (2022) reported for their own sensitivity test.

The same A/B on HD 209458 b. The `examples/15_molecular` configuration (metals off, `Base IR field: True` added, both runs restarted from its converged solution) reproduces the pattern on a real planet. Here the JFNK finish would not reach `Resid tol: 1.0e-5` with the band on: it stalled three times at $\|R\| = 1.5$–6.8×10^{-5} on the base-adjacent momentum row, open item (A), so *both* members of the pair were run at `Resid tol: 2.0e-5`, where both return `info = 0` and both mass fluxes are flat to 4.1×10^{-3} over $r > 1.2$.

	LW off	**LW on (343)**
residual norm at exit	1.77×10^{-5}	1.96×10^{-5}
$H_2 \to$H front ($x_{H_2} = 0.5$)	1.01578	**1.01290**
x_{H_2} at the base	0.99628	0.99605
q_{H_2} at the base	0.8512	0.8509
x_{H_2} at $r = 1.02$	2.37×10^{-4}	$\mathbf{4.22 \times 10^{-6}}$
H_3^+ peak [cm^{-3}]	4.81×10^5 @ 1.0010	5.46×10^5 @ 1.0010
T at $r = 1.005/1.02$	937.7 / 986.3 K	792.7 / 1275.8 K
$\log_{10}\dot{M}$, spherical	10.5711	10.5600

The base is again untouched ($N_{H_2} = 2.6 \times 10^{21}$ cm^{-2}, $f_{\rm shield} = 2.1 \times 10^{-6}$), the front moves in by $0.0029\,R_{\rm p}$, the H_2 fraction just above it falls by 56×, and $\dot{M}$ moves by −0.011 dex. The chemical-equilibrium fit gives $q_{H_2} = 0.831$ at this base and photochemistry would give less still; the network sits at 0.851 with the band on as without it.

10. The molecular infrared bands (`Molecular IR bands`, 2026-08-31)

The `Base IR field` closure of §8. gave the infrared coolants the code already had (the eight ground-term fine-structure lines and the H_3^+ bands) the field they sit in, and stopped there. Its own closing paragraph named what was left: past the $H_2 \to$H front there are no molecular coolants in the model at all, and holding that gas needs the coolants a real H_2 atmosphere carries. This section supplies three of them, and addresses open item (G).

Which coolants, and why these three. **H_2 quadrupole and magnetic dipole lines** (Roueff et al. 2019, table 2, VizieR J/A+A/630/A58). H_2 is the bulk gas: its transition probabilities are 10^{-10}–10^{-6} s^{-1}, but it outnumbers every other molecule by 10^3–10^4, and the two cancel. At $n(H_2) = 5 \times 10^{13}$ cm^{-3} and 1000 K it radiates 1.4×10^{-7} erg cm^{-3} s^{-1}, four times the local photoheating (the pure rotational part alone is 4.8×10^{-8}; above about 600 K the vibrational bands carry most of it). It is the only one of the three that exists in a Tier-2 run without the oxygen option. **H_2O** (HITEMP 2010) is the strongest infrared coolant of a solar-composition H_2 atmosphere below ∼2000 K, and the one the oxygen option switches the [O I] fine-structure lines off in favour of. **CO** (HITEMP 2019) holds essentially all the carbon at the base; its 4.7 μm fundamental carries four to five orders of magnitude more than its pure rotational lines above 300 K.
H_2–H_2 and H_2–He collision-induced absorption is deliberately *not* a channel, and that is a measurement rather than a scope decision. It scales as n^2, and on the CIA tables shipped with Photochem its optical depth over a pressure scale height at the 1 μbar base is 8×10^{-9} (hot Uranus) to 2×10^{-9} (HD 189733 b), with unit optical depth reached only near 0.2–0.4 bar: consistent with Lavvas & Arfaux (2021), who put CIA's onset at $p > 1$ bar. It is what makes the reservoir *below* the base black in the windows between the bands (the assumption `Base IR field` already encodes), not a local coolant of the modeled domain.

The closure. For each species the net radiative loss per unit volume is the optically thin, LTE exchange with the diluted field of the lower atmosphere,

$$\Lambda_{\rm net} = n_X \left[E_X(T) - W\, E_X^{\rm abs}(T; T_0) \right], \tag{13}$$

with, for a band absorber,

$$E_X(T) = 4\pi \sum_b \sigma_b(T) \int_b B_\nu(T)\, d\nu, \qquad E_X^{\rm abs}(T;T_0) = 4\pi \sum_b \sigma_b(T) \int_b B_\nu(T_0)\, d\nu, \tag{14}$$

and, for the H_2 line sum, the same quantity assembled transition by transition with the full two-level net factor $1+\bar{n}-\bar{n}\,e^{\Delta E/T}$. W is the dilution the H_3^+ closure uses, $\frac{1}{2}[1-\sqrt{1-(R_{\rm p}/r)^2}]$, and it is zero when `Base IR field` is off.

Keeping the stimulated-emission term is not decoration: it is what makes the bracket vanish identically at $T = T_0$, $W = 1$, gas buried in a blackbody at its own temperature neither cools nor heats. For the band absorbers that property is automatic, because the HITRAN line intensities the cross sections are built from already carry the $1-e^{-h\nu/kT}$ factor, so σ_b is the coefficient that pairs with B_ν. For a line sum it has to be written down. The H_3^+ closure of §8. had dropped it; it was put back the same day, and because $E_k(\nu_2)/T_0$ is large that cost only 2.2% of the emission and 5 K of equilibrium temperature there (936 → 941 K at $T_0 = 1140$ K). On a far-infrared band the same omission would be an order-unity error. The single-band H_3^+ form the 941 K belongs to was itself replaced the same day (a total emission fit cannot be paired with the Boltzmann factor of one transition at all temperatures), and the H_3^+ column below is the replacement's (§112 of `docs/Update_EXHALE_stage1.pdf`).

Each channel therefore has its own radiative equilibrium temperature, and it is the fixed point rather than the magnitude that answers item (G):

T_0	H_2O	CO	H_2	(H_3^+)
900 K	705 K	752 K	783 K	775.7 K
1140 K	889 K	918 K	984 K	958.5 K
1183 K	922 K	947 K	1019 K	991.0 K
1450 K	1120 K	1120 K	1226 K	1191.4 K
1800 K	1369 K	1338 K	1486 K	1451.6 K
2358 K	1749 K	1668 K	1883 K	1857.0 K

(half-sky coverage, $W = \frac{1}{2}$; the H_3^+ column solves $W_{\rm MiZATeP}(T_{\rm eq}) = \frac{1}{2}W_{\rm MiZATeP}(T_0)$ on the table and interpolation of `h3p_lte_emission_mizatep`, recomputed 2026-10-10 after the cooling function changed; with the Miller et al. 2013 fit it read 780.9, 975.4, 1010.7, 1231.6, 1517.6 and 1938.6 K). How much rests on that assumption is measurable: for H_2O at $T_0 = 1183$ K the equilibrium temperature is 726.5 / 922.1 / 1065.0 / 1183.0 K at $W = 0.25/0.5/0.75/1$, and CO 784.7 / 946.5 / 1073.1 / 1183.0 K, at $W = 1$ both are exactly T_0, the closure's fixed point recovered numerically. Because W multiplies only the absorption, these temperatures are insensitive to first order to any grey escape probability applied to both terms, which is why the optically thin limit is enough to fix the *temperature* even where a band is marginally thick.

Optical depth, and why the thin limit is used. At a 1 μbar base with solar oxygen the band-mean optical depth of the H_2O rotational bands over a scale height is a few times 10^{-2}; the strongest g-points inside a band reach order unity. The H_2 lines are thinner still: the line-centre opacity of 0–0 S(1) at 17 μm gives $\tau \approx 3\times10^{-4}$ over a scale height at $n(H_2) = 5\times10^{13}$ cm^{-3} and 1000 K. No escape probability is applied, and the Planck-mean optical depths of the H_2O and CO columns to the top of the domain are written to `output/Cooling_breakdown.txt` so the assumption

is measured rather than asserted. `output/FUV_bands.txt` carries the same statement as a conservation check: the column-integrated absorbed power of the three bands against the incident infrared flux $W\sigma T_0^4$, with a warning if it is exceeded.

Where the numbers come from, and three independent checks. The band-mean cross sections are $\sum_g w_g k_g$ of the correlated-k tables, which is exactly the bin-mean cross section. Line strengths do not depend on the broadening, only the line shape does, so this mean is pressure independent to better than 2% from 10^{-6} to 1 bar (measured), and the whole EXHALE domain uses the 10^{-6} bar slice, which is also the lowest tabulated pressure. The tables run 50–2000 K and are clamped outside; the H_2 line sum has no such limit. Three checks, against sources that share none of that line processing:

- the Planck-mean H_2O cross section agrees with the independent table `aiolos/inputdata/1H2-16O_T400.aiopa` to within 11% over 600–2000 K (ratio 0.889–1.111 on the 14 shared grid temperatures); below 600 K that table is not monotonic in T and the comparison is not meaningful, the low-temperature end being covered by the Neufeld & Kaufman comparison instead;
- restricted to $\lambda > 9\,\mu$m, i.e. to the pure rotational band, the H_2O emission agrees with the optically thin LTE rate of Neufeld & Kaufman (1993) table 2 to a factor 0.70 (100 K), 0.81 (200 K), 0.90 (400 K), 1.02 (1000 K); the same cut on CO against their table 3 gives 0.81–0.93 over 300–2000 K. At 100 K the CO ratio falls to 0.10, most likely the eight-point g-quadrature resolving the linear bin mean poorly for CO's sparse rotational ladder rather than the coarse long-wavelength binning (97% of the 100 K emission lands in the eighteen 9–250 μm bins); CO radiates 1.6×10^{-20} erg s^{-1} per molecule there against 1.8×10^{-15} for H_2O, so it does not matter;
- the H_2 line sum agrees with the LTE rates of Hollenbach & McKee (1979) eqs. (6.37) and (6.38) to 1–11% over 200–2000 K.

Two numbers from those comparisons are the reason the whole band system is integrated instead of a published *rotational* cooling function being adopted: above 300 K CO's total emission is four to five orders of magnitude above its rotational part, and H_2O's is 3.2× its rotational part at 1000 K. Neufeld & Kaufman's tables also stop at an optical depth parameter of 10^{19} cm^{-2} (km/s)$^{-1}$, a ceiling set by hydrostatic equilibrium in a self-gravitating cloud, which does not apply to a planetary atmosphere and which a base column exceeds.

What it does. Four converged solutions of the hot-Uranus gate (`data_g2` restarted with `Load IC? True`, `Solver: Newton 5.0e-2`, `Resid tol: 1.0e-5`, `Max steps: 150000`), all with `Base IR field: True`, so each pair differs only in `Molecular IR bands`. This configuration has no oxygen chemistry, so the only new channel is the H_2 line sum. The bands-off members reproduce the field-on solutions of §8. to 1–3 K (metals on) and 5–30 K (metals off, the difference being the H_3^+ stimulated-emission fix).

	metals OFF		metals ON	
	off	**on**	off	**on**
JFNK iterations	89	**21**	25	26
residual norm at exit	8.77e-6	8.39e-6	2.00e-6	2.82e-6
$T(r = 1.005)$	940.5 K	912.1 K	1097.0 K	1044.1 K
$T(r = 1.02)$	878.6 K	857.6 K	926.1 K	887.0 K
$T(r = 1.05)$	750.3 K	721.9 K	572.2 K	588.7 K
$T(r = 1.08)$	420.9 K	345.4 K	215.1 K	236.1 K
coldest cell below 1.3	245.6 K @1.0972	225.4 K @1.0923	190.4 K @1.0842	198.2 K @1.0853
$H_2 \rightarrow$H front	1.08817	1.08532	1.07916	1.07970
$\log_{10}\dot{M}$ reported	10.32	10.31	10.29	10.30
mass-flux spread, $r > 1.2$	2.39e-3	2.38e-3	2.38e-3	2.38e-3

The solution's own zero crossing lands on the predicted equilibrium temperature: in the metals-on solution the H_2 channel changes sign at $r = 1.0122$, where the gas is at 951.7 K and the sky dilution is $W = 0.423$, and the closure evaluated at that dilution gives 951.5 K; the second crossing ($r = 1.1239$, $T = 850.6$ K, $W = 0.272$) is predicted at 872.7 K. Nothing was fitted.

The H_2 channel is *first order and does not repair the collapse*, and those are not in tension. In the metals-on solution H_2 carries 56% of the whole radiative rate at $r = 1.002$–1.005 (cooling there, since $T = 1044$–1124 K is above its radiative equilibrium temperature), and essentially all of the net at $r = 1.02$–1.05, heating. The layer's radiative time falls with it: $t_{\rm rad}/t_{\rm flow}$ at $r = 1.05$ goes from **8.22 to 0.263**. But the coldest cell sits *at or beyond* the $H_2 \rightarrow$H front in every run, where there is no H_2 to absorb anything and $t_{\rm rad}/t_{\rm flow}$ is 0.7–1.9; that gas is expansion-dominated and no molecular coolant can reach it.

With the oxygen chemistry on, where H_2O and CO exist, the magnitudes differ by five orders: on HD 189733 b the two bands carry 4.8×10^{-5} and 2.7×10^{-5} erg cm^{-3} s^{-1} at $r = 1.005$ against a total of 7.5×10^{-5}, and a 12000-step A/B moves the base from 760.2 to 829.1 K and $x_{\rm H_2}$ from 0.330 to 0.483, both toward the photochemical reference (864 K, 0.910). Neither of those runs is converged; `md/Update_EXHALE_stage1.md` §110 carries the full statement and its caveats.

What is assumed. LTE level populations (the rotational levels thermalize far below the densities of this layer; $n_{\rm crit}(H_2\, v{=}1)$ is 10^9–10^{11} cm^{-3} against 10^{13}–10^{14} cm^{-3} at the base, and the vibrational bands are only 0.7% of the H_2 emission at 400 K, 5.7% at 500 K and 18% at 600 K, so the overestimate is confined to the warm, dense part of the layer). The incident field is not attenuated by the intervening molecular column of the domain itself, the same approximation `Base IR field` makes for H_3^+. OH is left out: it has no cross-section table here and carries a few percent of the oxygen where the water carries most of it. The atomic advection post-process (`T_equation`) does not carry these channels, exactly as it does not carry H_3^+.

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